

VOLUME 13

Development Roadmap & Sprint Plan

Part 1: Master Implementation Roadmap

Project: Bidra Platform

Version: 1.0

Purpose

This document defines the official implementation roadmap for the Bidra platform.

Its objectives are to:

- Define the recommended implementation order.
- Minimize technical dependencies.
- Reduce rework.
- Allow parallel development where possible.
- Provide a clear roadmap for engineers and AI coding agents.
- Ensure every sprint produces a deployable increment.

This roadmap assumes all previous specification volumes have been completed and approved.

Development Philosophy

Bidra should be built from the **inside out**.

The order of implementation should always be:

1. Foundation
2. Security
3. Core Domain
4. Financial Operations
5. User Experience
6. Intelligence
7. Production Hardening

This ensures every new module builds on stable, reusable foundations.

Delivery Strategy

The platform will be delivered in **11 implementation phases**.

Each phase produces a fully integrated, testable increment.

No phase should begin until the exit criteria of the previous phase have been satisfied.

Phase 0 – Platform Foundation

Objective

Build the technical foundation required by every future module.

Deliverables

- Monorepo
- Next.js
- NestJS
- PostgreSQL
- Prisma
- Redis
- BullMQ
- Docker
- CI/CD
- Logging
- Health Checks
- OpenAPI
- Shared UI Library
- Shared Configuration
- Environment Management
- Error Handling
- Testing Infrastructure

Dependencies

None.

Exit Criteria

- Applications compile.
- Docker environment starts successfully.
- CI pipeline passes.

- OpenAPI is generated.
- Health checks pass.
- Shared packages are operational.

Complexity

Medium

Phase 1 – Authentication & Identity

Objective

Implement platform identity and access management.

Deliverables

- Registration
- Login
- Logout
- Password Reset
- Email Verification
- Sessions
- JWT
- Refresh Tokens
- Roles
- Permissions
- Organization Membership
- Audit Logging

Depends On

Phase 0

Exit Criteria

- Authentication is production-ready.
- Role-based authorization works.
- Session management is operational.
- Security tests pass.

Complexity

High

Phase 2 – Organizations

Objective

Implement organizations and verification.

Deliverables

- Organization Profiles
- Verification
- Branding
- Membership
- Public Profiles
- Organization Dashboard
- QR Codes
- Search

Depends On

- Phase 1

Exit Criteria

- Organizations can register.
- Verification workflow works.
- Public profiles are operational.
- Organization dashboard functions correctly.

Complexity

Medium

Phase 3 – Campaigns & Needs

Objective

Implement Bidra's primary business domain.

Deliverables

- Campaigns
- Needs
- Categories
- Updates

- Public Pages
- Search
- QR Codes
- Progress Tracking

Depends On

- Phase 2

Exit Criteria

- Organizations publish campaigns.
- Needs are publicly accessible.
- Progress calculations work.
- Search indexes correctly.

Complexity

High

Phase 4 – Contributions

Objective

Implement every contribution type.

Deliverables

- Financial Contributions
- Volunteer Contributions
- Goods
- Skills
- Equipment
- Sponsorship
- Awareness
- Contribution Timeline
- Verification
- Contribution Dashboard

Depends On

- Phase 3

Exit Criteria

- Every contribution type works.

- Need progress updates correctly.
- Contribution workflows are verified.

Complexity

Very High

Phase 5 – Financial Operations

Objective

Implement the complete payment ecosystem.

Deliverables

- Stripe Connect
- Checkout
- Payments
- Refunds
- Transfers
- Payouts
- Financial Ledger
- Receipts
- Reconciliation

Depends On

- Phase 4

Exit Criteria

- Payments complete successfully.
- Ledger balances.
- Refunds work.
- Webhooks are idempotent.
- Financial reports reconcile.

Complexity

Very High

Phase 6 – Volunteer & Recognition

Objective

Implement volunteering and Time Credits.

Deliverables

- Volunteer Opportunities
- Attendance
- Verification
- Time Credit Ledger
- Wallet
- Recognition Partners
- Rewards
- Voucher Redemption
- QR Validation

Depends On

- Phase 4

Exit Criteria

- Volunteers earn Time Credits.
- Rewards redeem correctly.
- Ledger remains balanced.
- QR validation prevents duplicate redemption.

Complexity

High

Phase 7 – Administration

Objective

Implement the operational control center.

Deliverables

- Admin Dashboard
- Moderation
- Platform Settings
- Trust & Safety
- Feature Flags
- Audit Viewer

- Financial Administration

Depends On

- Phases 1–6

Exit Criteria

- Platform administration is complete.
- Permissions function correctly.
- Audit logging is comprehensive.

Complexity

High

Phase 8 – Reporting & Analytics

Objective

Build the business intelligence layer.

Deliverables

- Dashboards
- Reports
- KPIs
- Scheduled Reports
- Exports
- Analytics Engine

Depends On

- Phases 1–7

Exit Criteria

- Dashboards reconcile with transactional data.
- Reports export successfully.
- Analytics performance meets targets.

Complexity

Medium

Phase 9 – Production Hardening

Objective

Prepare the platform for public launch.

Deliverables

- Performance Optimization
- Security Review
- Accessibility Audit
- Load Testing
- Disaster Recovery
- Backup Verification
- Monitoring
- Alerting

Depends On

- Phases 0–8

Exit Criteria

- Security review completed.
- Performance targets met.
- Accessibility validated.
- Backup and recovery tested.

Complexity

High

Phase 10 – Launch Readiness

Objective

Complete final validation before production deployment.

Deliverables

- Final QA
- User Acceptance Testing

- Documentation Review
- Release Notes
- Production Configuration
- Deployment Verification
- Go-Live Checklist

Depends On

- Phase 9

Exit Criteria

- All acceptance tests pass.
- Documentation is complete.
- Rollback plan validated.
- Platform approved for production release.

Complexity

Medium

Parallel Development Opportunities

The following workstreams may proceed in parallel after their dependencies are satisfied:

Workstream	Can Start After
UI Components	Phase 0
Design System	Phase 0
Authentication UI	Phase 1
Organization UI	Phase 2
Campaign UI	Phase 3
Payment UI	Phase 5
Volunteer UI	Phase 6
Reporting UI	Phase 8

Definition of Phase Completion

A phase is complete only when:

- All planned functionality is implemented.
- All acceptance criteria are met.
- Automated tests pass.

- Documentation is updated.
 - OpenAPI documentation is current.
 - Database migrations are complete.
 - Security review has been performed.
 - Performance is acceptable.
 - Code review has been completed.
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AI Execution Rules

For every phase:

1. Review the relevant specification volumes.
2. Create an implementation plan.
3. Build backend.
4. Build frontend.
5. Build tests.
6. Update documentation.
7. Perform self-review.
8. Resolve identified issues before proceeding.

Never begin the next phase until the current phase satisfies its exit criteria.

Release Milestones

Milestone	Description
M1	Foundation Complete
M2	Secure Identity Platform
M3	Organizations Live
M4	Campaign MVP
M5	Contribution Engine Complete
M6	Financial Platform Operational
M7	Volunteer Ecosystem Complete
M8	Administration Complete
M9	Analytics Complete
M10	Production Ready
M11	Public Launch

Final Principle

The roadmap exists to maximize delivery quality rather than delivery speed.

Every phase should leave the platform in a stable, deployable state.

Progress is measured by completed, production-ready functionality, not by the amount of code written.